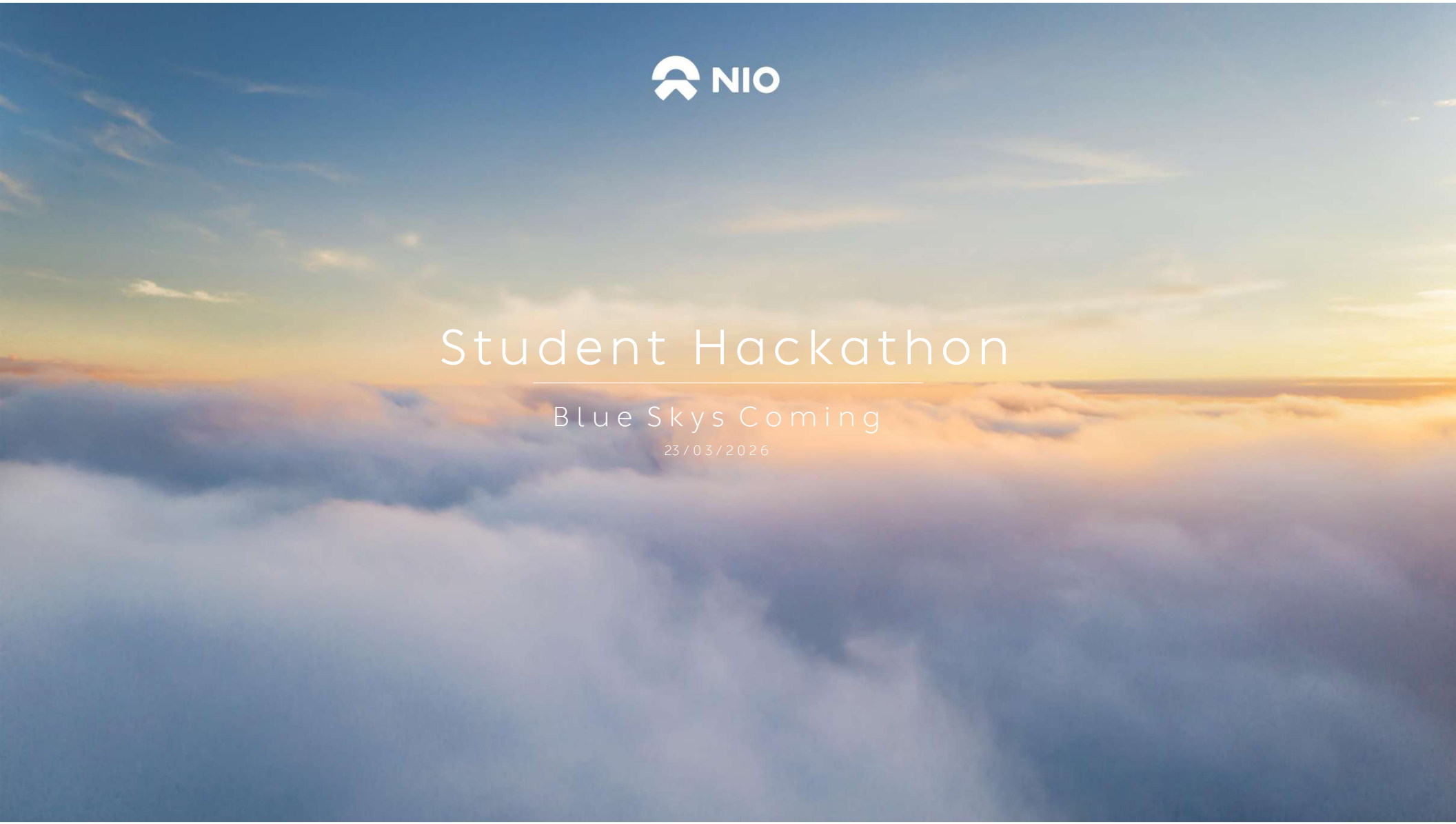
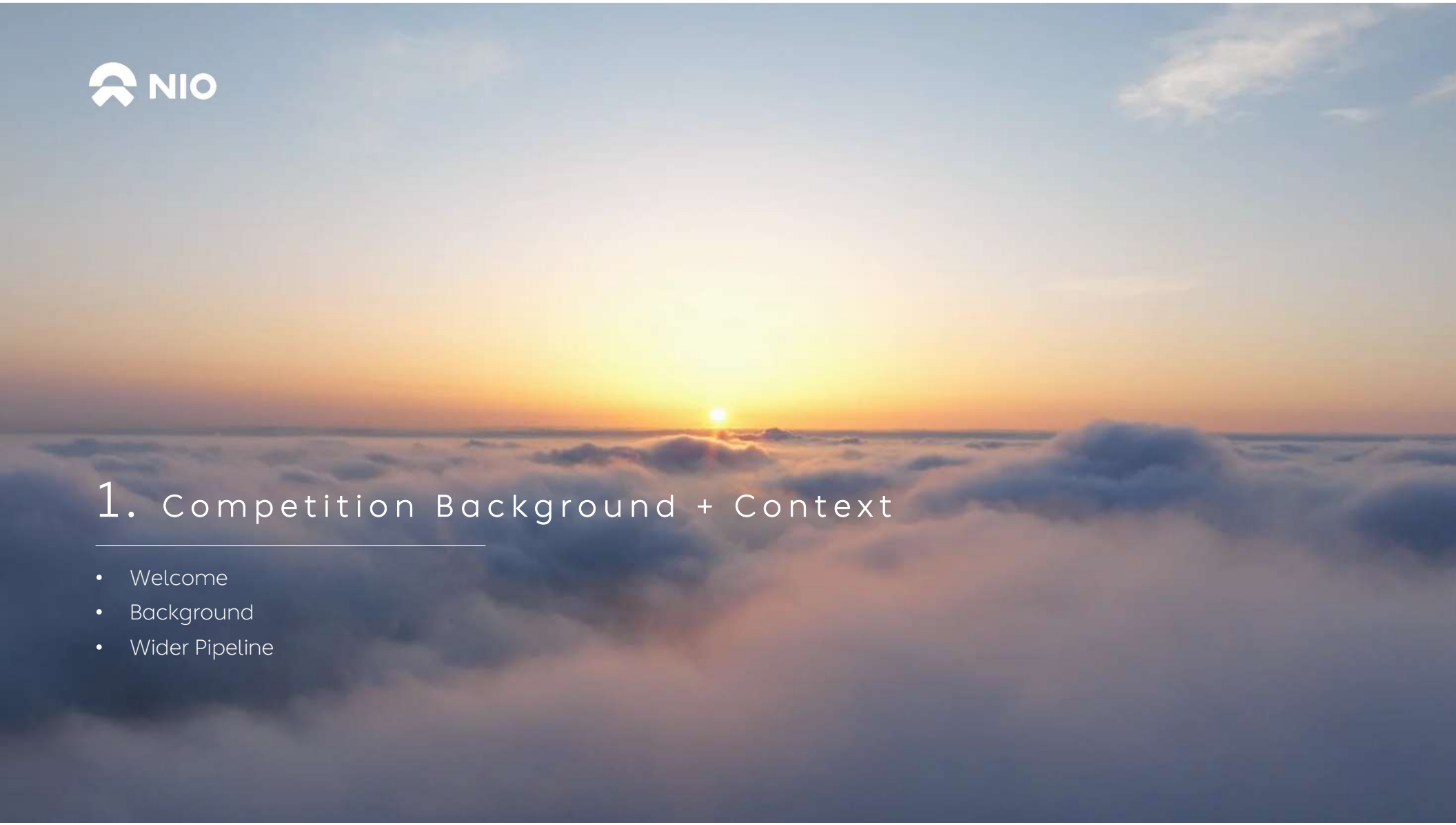


NIO

Student Hackathon

Blue Skys Coming

23/03/2026



1. Competition Background + Context

- Welcome
- Background
- Wider Pipeline

01 . 1

Welcome

Thank you for registering for the event.

The NIO Student Hackathon is a technical data challenge inviting students across UK universities to solve a real-world engineering problem at the intersection of automotive engineering, computer vision, and AI-driven quality assurance.

Participants will work in teams to develop an automated Python solution to an open problem, contributing towards research into scalable quality assurance processes within advanced engineering simulation pipelines.

The winning team will receive a cash prize of £4,000 split evenly among team members.

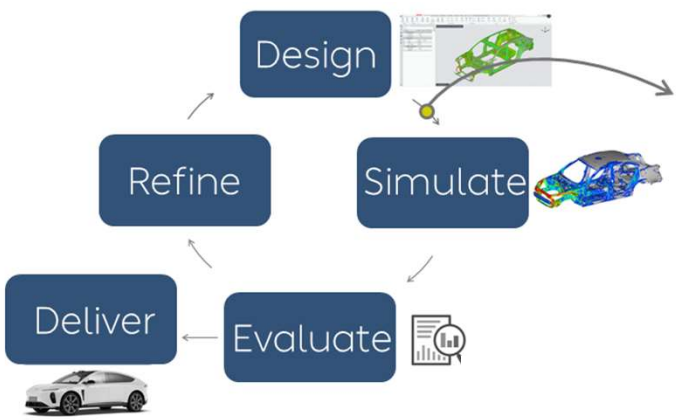
A unique opportunity to engage in a real-world data challenge at one of the world's leading EV companies.

01. 2

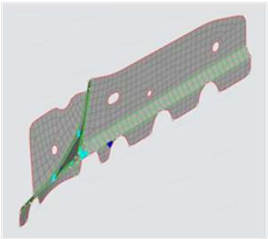
Background

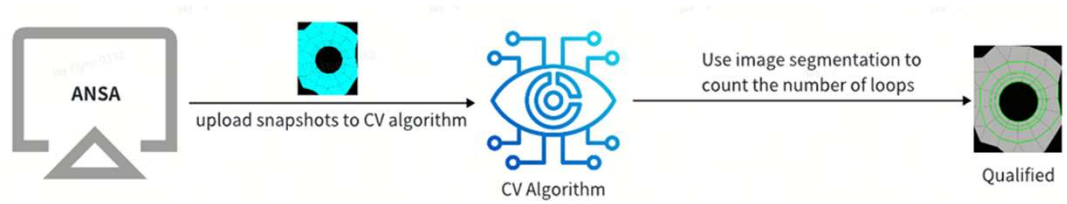
- NIO uses an iterative CAD refinement process that includes automated quality checks before simulation (e.g. crash testing)
- A key quality check ensures bolt/fixing holes are correctly meshed, currently handled by bespoke computer vision scripts
- The hackathon aims to develop an improved method using CNNs or Multi-modal LLMs to improve on current methods
- The main challenge is insufficient labelled training and testing data, particularly for failure cases.
- This hackathon aims to develop methods to address this challenge, by solving a data challenge to contribute towards a synthetic data generation pipeline

High-Level Automotive CAD Simulation Pipeline



QA checks are performed to ensure mesh objects meet the required quality standards for simulation.





01. 3

Background

This hackathon addresses the first step in a larger research project.

This event is also part of a wider talent scouting initiative for NIOs grad schemes, industry funded projects, and other academic collaborations.

Further information on the project background and context can be found in the pre-registration pack.

For more details, please scan the QR code below:

or Click Here

1 .jpeg to .nas file conversion (Hackathon Topic)

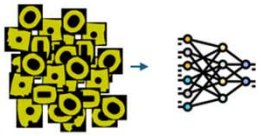
Import .jpeg images → Pre-process images → Extract Node Coordinates → Identify element Nodes → Output nodes & elements in .nas format

2 .nas file manipulation to generation high volume training dataset

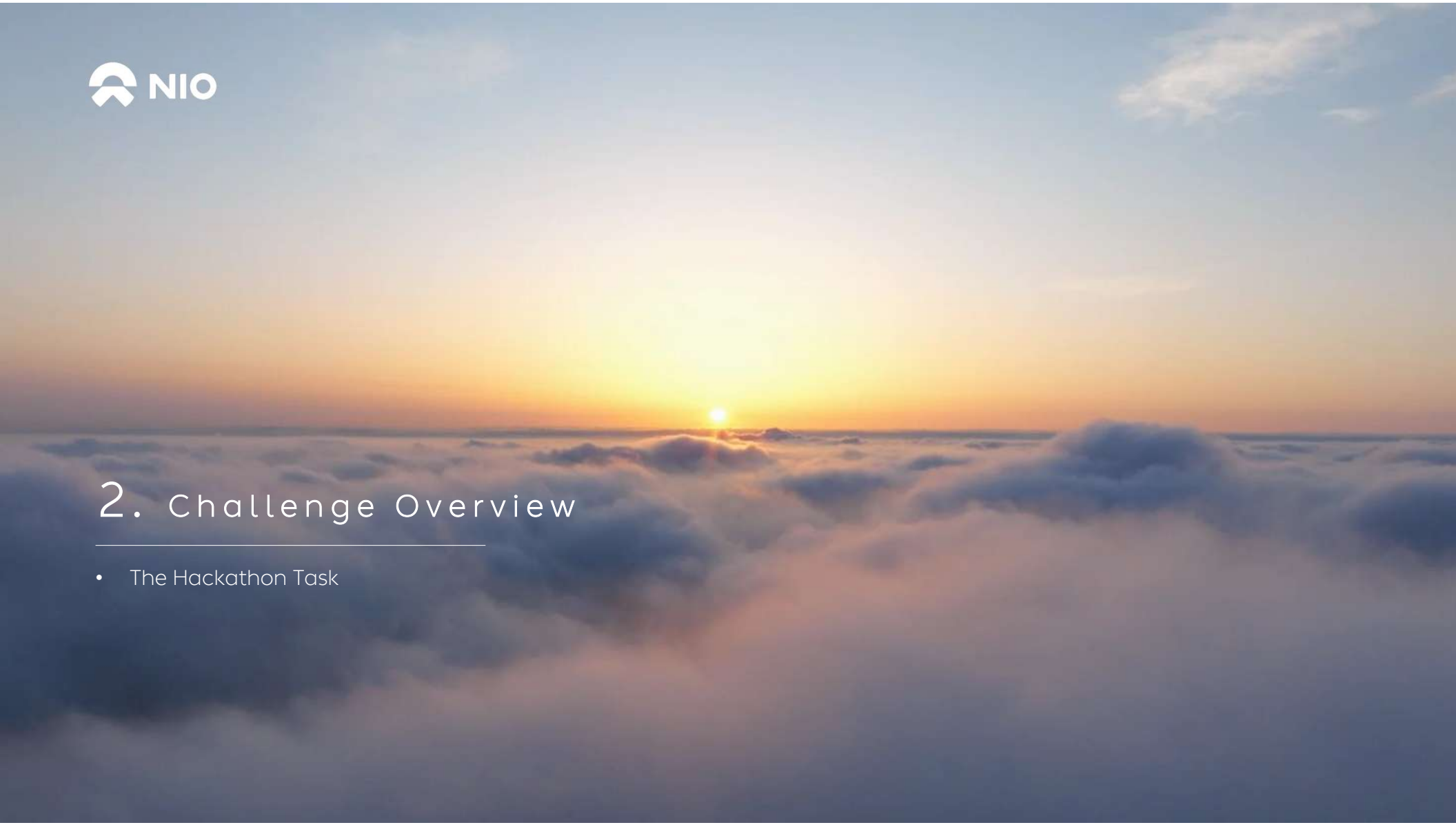
Import .nas file → Modify mesh to augment image → Reproduce this process to create very large dataset

3 Train CNN, Multi-modal LLM , or similar approaches, with new expanded dataset

Train using synthetic images → Test using original images → Compare range of CNNs & other methods → Evaluate performance



4 Explore opportunities to replicate this approach in other QA checks.



2. Challenge Overview

- The Hackathon Task

02 . 1

Hackathon Summary

Objective

Participants are provided with a dataset of 100 .jpeg images representing finite element (FE) meshes.

The goal is to develop a Python-based solution that takes a .jpeg image as input and outputs a corresponding .nas file that details the underlying mesh geometry.

No prior experience with .nas files is expected. Full .nas format details are provided in this pack.

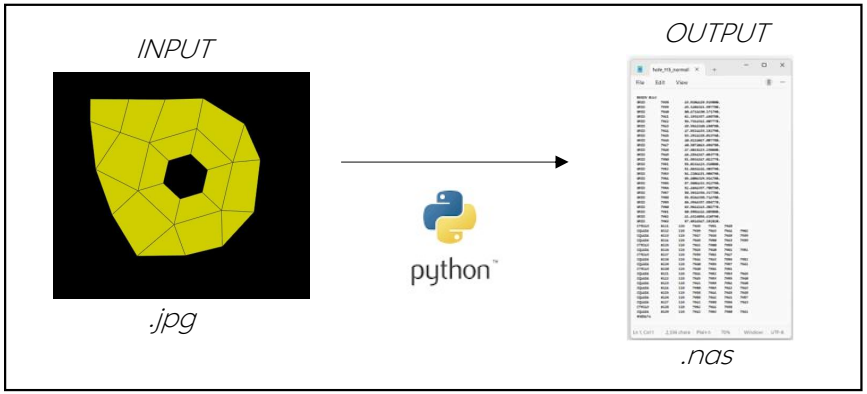
Required Submission

Each team must submit:

1. Generated output of the 100 .nas files corresponding to the provided .jpegs
2. The full, reproducible, well-documented code used to generate the .nas files
3. A concise one-page report that summarises the approach, methods, and tools used (including AI/LLMs, if applicable).

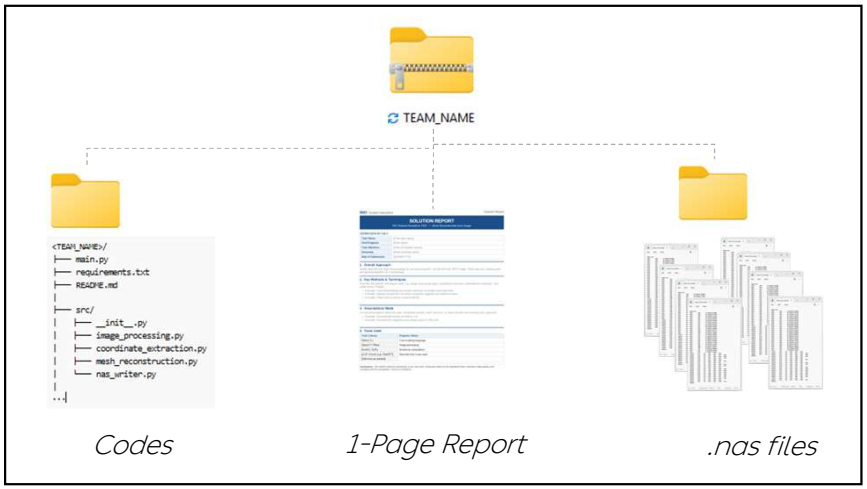
Task Overview:

Use Python to produce a .nas file from a .jpeg



Submission Overview:

.zip file containing: codes, 1-page report, .nas files



02 . 2

The Data

A GitHub repo contains:

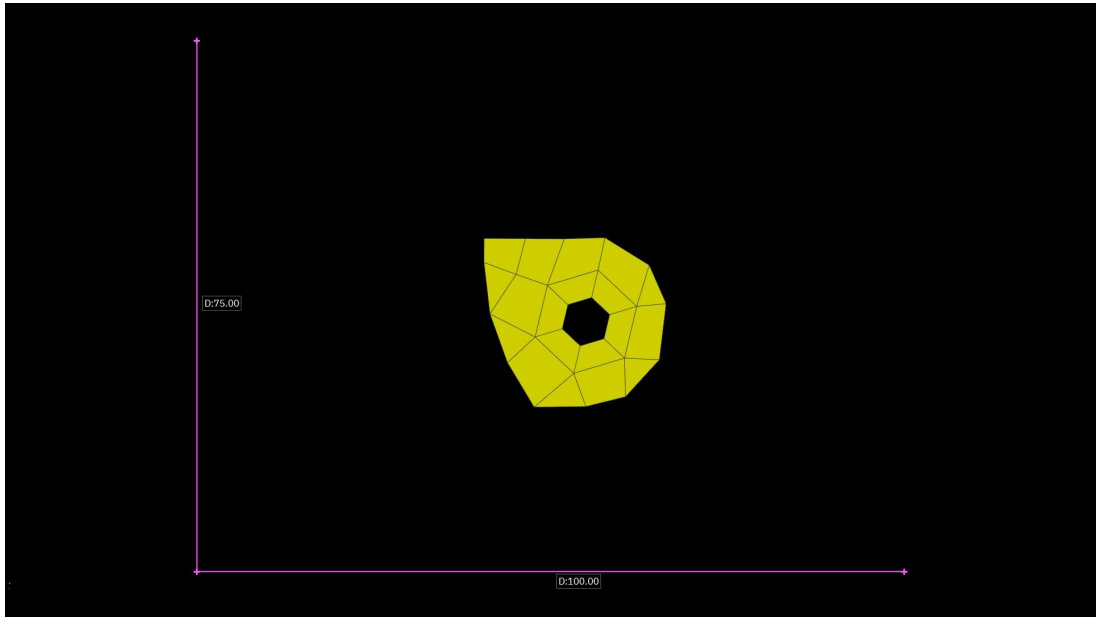
- 100 .jpeg images representing FE meshes.
- 2 example images representing FE meshes with corresponding .nas files

Use the example images and .nas files to familiarize yourself with the .nas format.

A simplified example of how a basic element is constructed using a .nas file is included on the following slide.

You will have received an invitation to the GitHub upon sign up. If you have signed up and not received a link, please contact: jay.flynn.o@nio.io

hole_115_normalized.jpg



The above image is one of the provided examples and includes a corresponding .nas file.

Coordinates System

Note the purple lines used to define the coordinate system for ALL images, where:

- the origin (0,0) in the bottom left +
- y_max = 75.00 at the top purple +
- x_max= 100.00 at the right purple +

.nas Format Overview

- *Note that colour, line style, node size, and other variable plot formats and styles are not considered in the .nas file.*
- *This is a simplified visual example of a .nas file. For exact format, see the 2 example .nas in the GitHub repo*

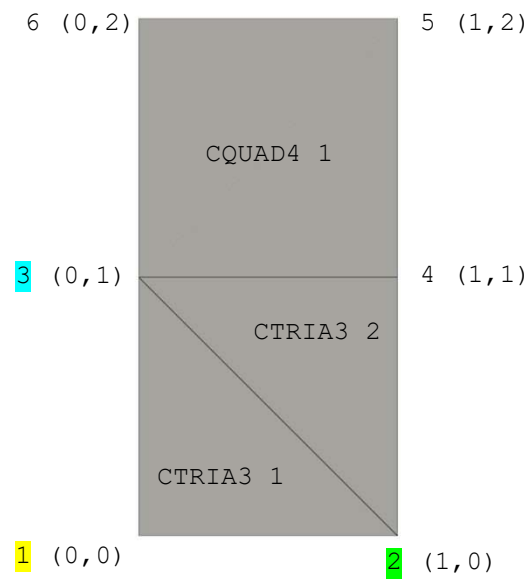
Coordinate ID

	BEGIN BULK			X	Y	Z	
Coordinates	GRID	1		0.000	0.000	0.000-1	
	GRID	2		1.000	0.000	0.000-1	
	GRID	3		0.000	1.000	0.000-1	
	GRID	4		1.000	1.000	0.000-1	
	GRID	5		1.000	2.000	0.000-1	
	GRID	6		0.000	2.000	0.000-1	
Elements	CTRIA3	1	90690	1	2	3	
	CTRIA3	2	90690	2	3	4	
	CQUAD4	3	90690	3	4	5	6
	ENDDATA						

Coordinate ID

Element ID

File Ref. No.

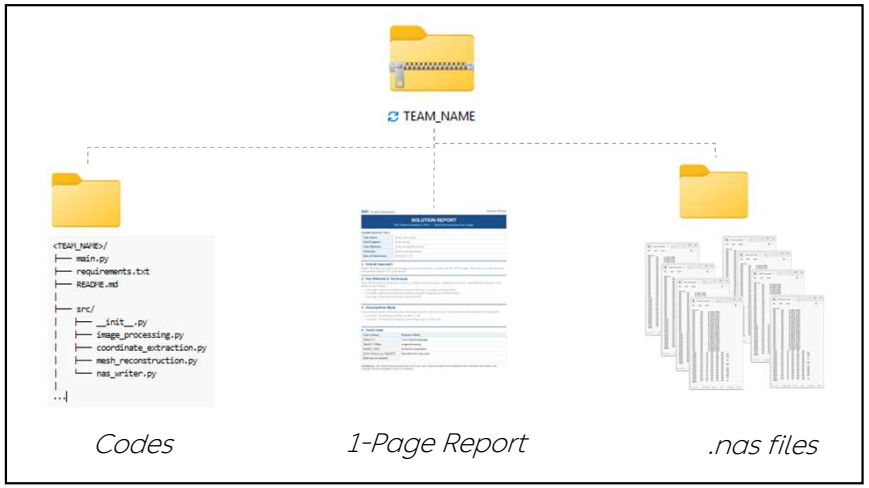


02 . 4

Submission Process

- Each team's chief engineer is responsible for the submission.
- To submit, the chief engineer will email a .zip file containing the teams submission to jay.flynn.o@nio.io before 23:59 on Friday 17th April.
- Late submissions will not be considered.
- Teams are only permitted to submit a max of 1 submission. Resubmissions will not be considered.
- The submission should follow the suggested format outlined in the GitHub Repo.
- If the file is too large to be shared via email, a link to download the file may be accepted. In this case, please liaise with jay.flynn.o@nio.io well ahead of the submission deadline to ensure the file is accessible by NIO prior to the deadline.
- Teams must abide by the T&Cs. If ANY team member is found to be in breach of any of the terms and conditions[1], they risk the entire team being disqualified.

Submission Overview:
.zip file containing: codes, 1-page report, .nas files



[1] Terms and Conditions - https://nio.jp.feishu.cn/docx/D1Q0dlw3KoOpoKxGRuQjgi53puf

02 . 5

Submission Documents

Each team's chief engineer must submit:

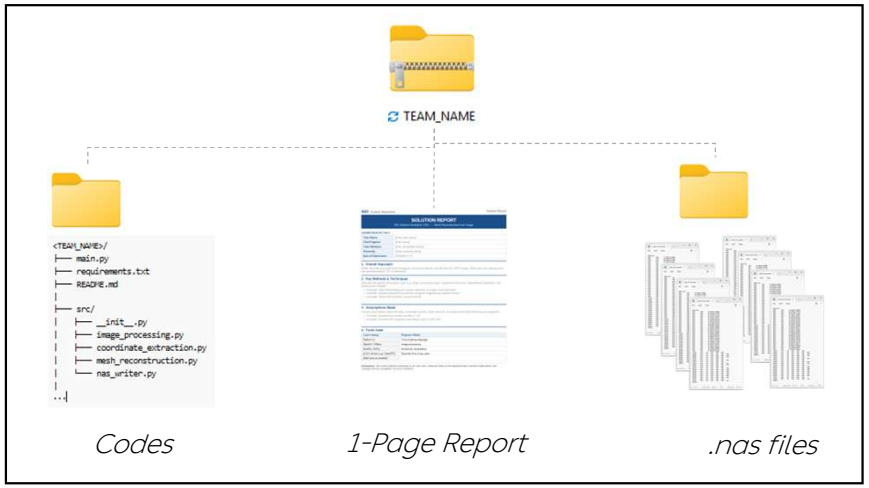
1. Generated output of the 100 .nas files corresponding to the provided .jpegs
    - .nas files must follow the example format
    - Each .nas file must have an identical name to the image for which it represents
    - Each .nas file must have the correct 'File Ref. No.'

2. The full code used to generate the .nas files
    - The provided .nas files MUST be reproducible using this code
    - Codes should be executed by running a main.py file
    - Codes should be well-documented throughout, with clear instructions on how to use the code to reproduce the .nas files
    - A clear, well formatted README.md file should be included

3. A concise, well presented one-page report that summarises the approach, methods, and tools used (including AI/LLMs, if applicable).
    - .docx or .pdf format

Omitting any of these requirements may result in your submission being rejected.

Submission Overview:
.zip file containing: codes, 1-page report, .nas files



02 . 6

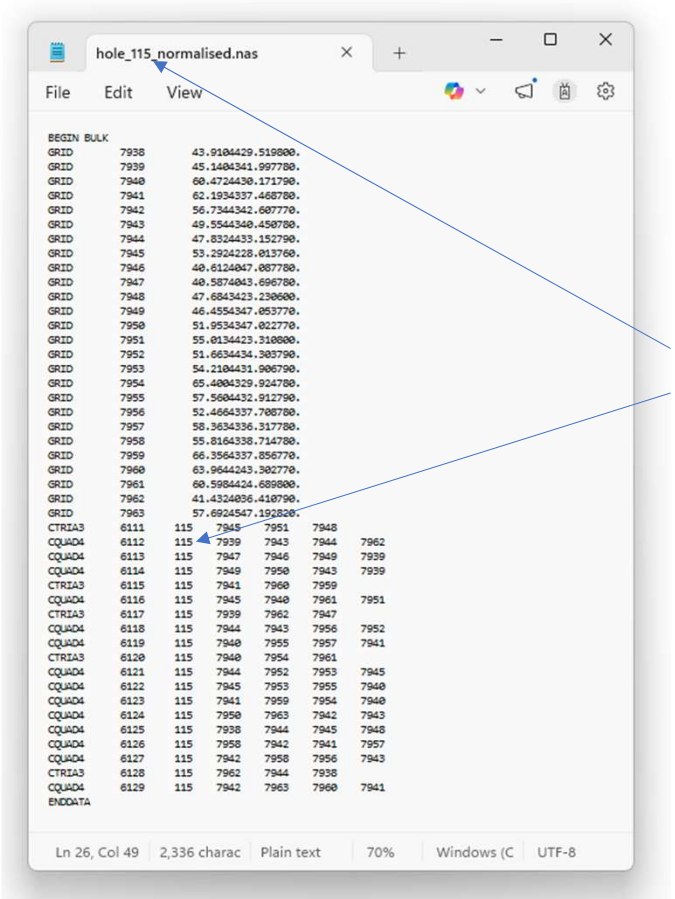
Scoring

Provided all T&Cs and submission requirements are met, teams are primarily scored based on the accuracy of 100 .nas submitted.

Each .nas file is scored against the original ground truth .nas that was used to produce the image.

Each .nas file is scored out of 100. scoring is performed as follows:

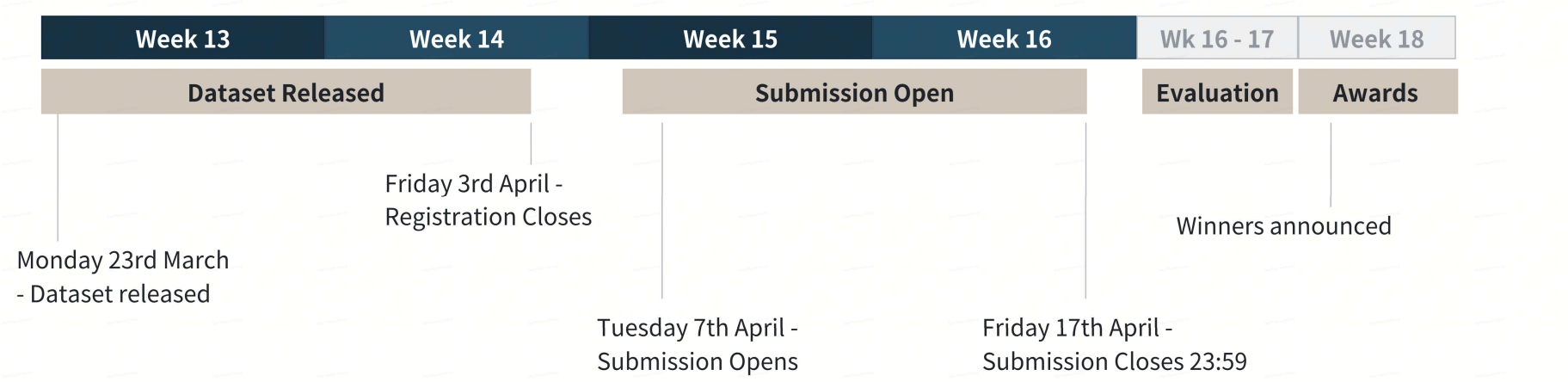
1. The format must match the provided example format
2. The 'BEGIN BULK' and 'ENDDATA' strings must be included.
3. The number of total GRID rows must exactly match the ground truth.
4. The total number of TRID and QUAD elements must exactly match the ground truth
5. For each element, the GRID coordinates defining the geometry must exactly match the ground truth
6. For each .nas file, if conditions 1-5 are met the score is calculated by measuring the sum of average distance from each GRID coordinate to its respective ground truth coordinate.
7. If conditions 1-5 are not met for any given .nas, that file will score 0.
8. The total score for a team is the average score across all 100 .nas files.
9. Any ties will be decided by assessing the quality of the code documentation and the 1-page report.

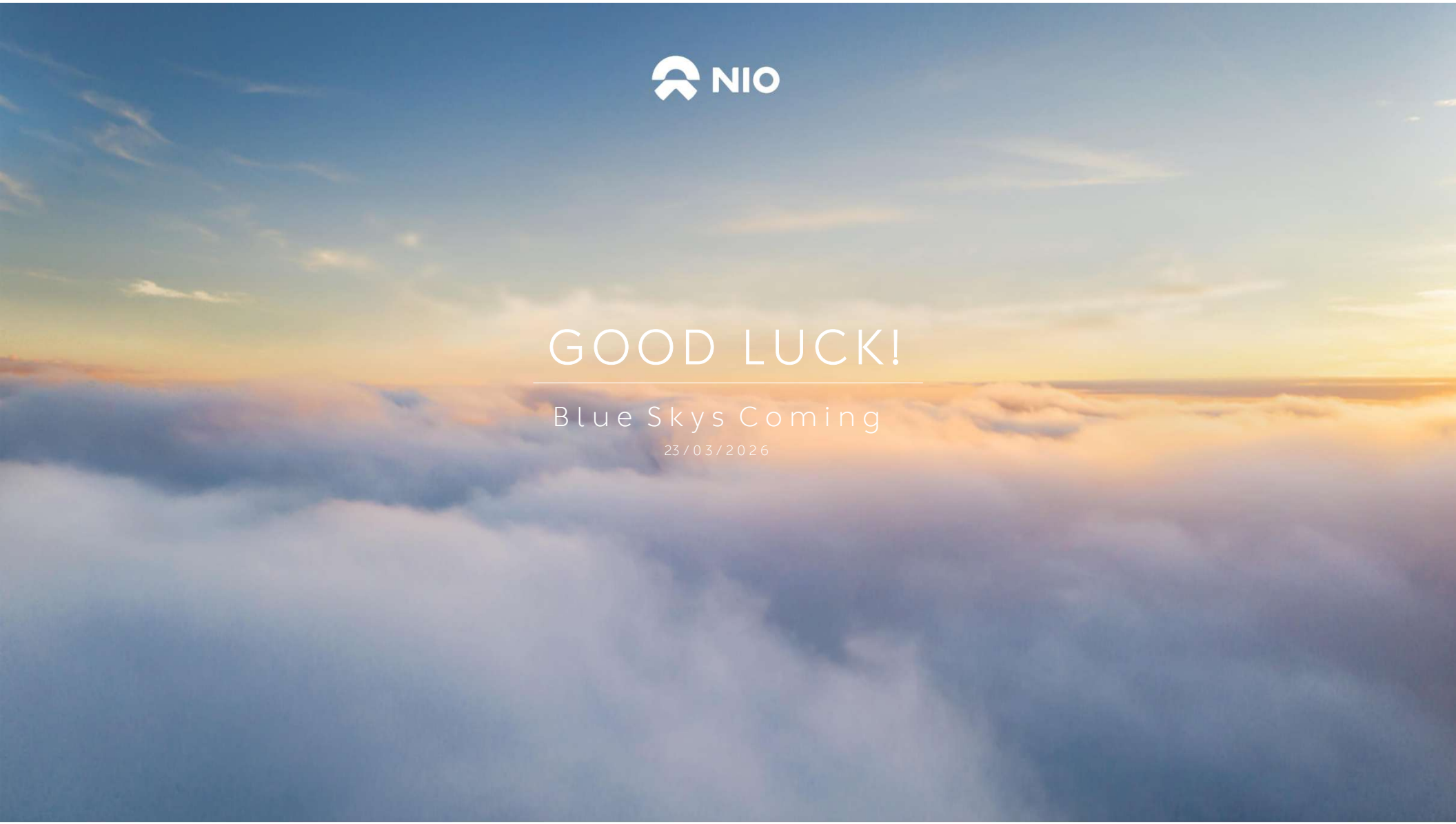


File name & 'File Ref. No.' must match

02 . 7

Competition Timeline




NIO
GOOD LUCK!
Blue Skys Coming
23/03/2026